

Entities become invisible under certain conditions

MC-65040 Entities

Status	Resolved / Fixed
Affected	Minecraft 14w31a, Minecraft 14w34b, Minecraft 1.8-pre1, Minecraft 1.8-pre2, Minecraft 1.8-pre3, Minecraft 1.8 (+62 more)
Fixed in	Minecraft 15w38b, Minecraft 15w39a, Minecraft 15w40a
Reported	2014-08-01

STEPS TO REPRODUCE

1. Launch Minecraft 18w43a and create a new Creative world, preferably superflat.
2. Open Options -> Video Settings and set Render Distance to 2.
3. At the exact point where you spawn, place an armor stand.
4. Travel exactly 100 blocks in one direction and place a second armor stand.
5. Travel exactly 100 blocks back to the first armor stand.
6. Use `/tp @e[type=ArmorStand,c=-1]` to confirm that the armor stands are still present.
7. Observe that the armor stands are completely invisible even though they remain present and can still be interacted with.

OBSERVED BEHAVIOUR

Actual behavior

After the player leaves entities in constantly loaded chunks and later returns, the entities may become completely invisible. This occurs most often with low render-distance settings, particularly 2-5 chunks, and is more frequent in 18w43a.

The entities remain present in the world and can often still:

- Produce sounds
- Be interacted with
- Be selected or teleported to using commands
- Be picked up or affected normally, in the case of XP orbs

Affected entities include mobs, XP orbs, boats, minecarts, armor stands, and empty or filled item frames. Dropped items on the ground do not appear to be affected. The problem is limited to entities being fully invisible; visual distortions or partial invisibility are not part of this issue.

The issue occurs when an entity remains loaded while outside the player's render distance, particularly in spawn chunks or other chunks kept loaded by chunk loaders or portals. Moving away from such an entity and then returning can cause it to stop rendering even though it is still present on the server. Leaving and re-entering the world causes the affected entities to render normally again.

EXPECTED BEHAVIOUR

Entities placed or loaded in always loaded chunks should remain fully visible when the player moves out of render distance and returns, including mobs, armor stands, item frames, boats, minecarts, and experience orbs. Changing the player's location or using a low render distance should not make entities disappear while they remain present and interactable, and they should not require leaving and re-entering the world to become visible again.

ENVIRONMENT

Minecraft Java Edition Versions: 14w31a, 14w34b, 1.8-pre1, 1.8-pre2, 1.8-pre3, 1.8, 1.8.1-pre1, 1.8.1-pre2, 1.8.1-pre5, 1.8.1, 1.8.2-pre1, 1.8.2-pre4, 1.8.2-pre6, 1.8.2, 1.8.3, 1.8.4, 1.8.6, 1.8.8, 15w32a, 15w32b, 15w33b, 15w33c, 15w34a, 15w36c, 15w36d, 15w37a, 15w45a, 15w46a, 15w47a, 1.8.9, 15w50a, 16w04a, 16w05b, 1.9 Pre-Release 1, 1.9 Pre-Release 4, 1.9, 1.9.4, 1.10, 1.10.2, 1.11, 1.11.2, 17w06a, 17w13a, 1.12, 1.12.1, 1.12.2, 1.13-pre3, 1.13-pre5, 1.13-pre7, 1.13, 1.13.1, 1.13.2, 18w43a, 18w43b, 18w43c, 18w44a, 18w45a, 18w46a, 18w47a, 18w47b, 18w48a, 18w48b, 18w49a, 18w50a, 19w02a, 19w07a, 19w08a, 19w08b (includes snapshots)

ORIGINAL DESCRIPTION

18w43a - This issue has now gotten worse in 18w43a, appears to happen a lot more frequently

This report only covers totally invisible mobs. (No distortions or partially invisible). Entities seem to become invisible in several ways in 1.8. In this report one of the ways are described. If any of these things listed below applies to what you are seeing, this is probably the right place, if not then you should probably look in the "related to" list below for the bug you are seeing:

- It is affected by draw distance.
- It happens only in or close to always loaded chunks (Spawn chunks or chunks loaded by chunkloaders or portals)
- It affects most entities, but not items just laying on the ground.

(Any the above alone is enough)

The reason that the other related bugs are not covered here is that they are not reproduced in the same way and they don't overlap much from a testing point of view, so it would be confusing to mix them together. However some of the other bugs around are probably heavily related to this one so in case of a fix, they should probably all be looked through. (Things turning invisible)

Note: Due to recent merge with

MC-66683

Resolved

and also that I got to take over this report, the description here does not yet cover everything. It also needs some cleaning up. This however will be fixed soon. Until then most of the important parts are copied in here, but be sure to have a look at that report if you need extra info. Just make sure to post comments here to get everything in the same place.

General info

- Mobs and most other entities become invisible, but will still make sounds and can in some cases be interacted with.
- The issue covered here is only about mobs becoming totally invisible under certain conditions. It does not cover distorted or partially invisible mobs.

Conditions for it to occur

- The entity needs to be loaded in the world while still be out of render distance and can never get unloaded, between the time you leave the area until you come back, for this to occur.
- It happens using lower render distance settings (2-5) with it being easier to reproduce the lower the render distance is set to.

MANUAL RESULT

Version used	
Reproduced?	YES / NO / PARTIAL
Crash file	
Notes	
Tested by / date	

Live issue: <https://bugs.mojang.com/browse/MC/issues/MC-65040>